2027-2031 Distribution Capex Plan

Business case for the recommended investment portfolio · scenario: **base** · analysis run #1 · 15 October 2026

PKR 1,648 mn Portfolio NPV (real, at plan start)	**PKR 2,675 mn** Capex committed of PKR 2,760 mn budget	**21.3%** Portfolio IRR (real)
1.68x Benefit-cost ratio	**100%** Probability portfolio NPV > 0	**18 / 38** Projects funded

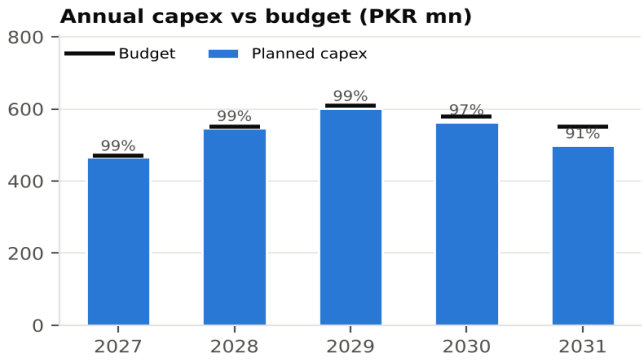
Recommendation

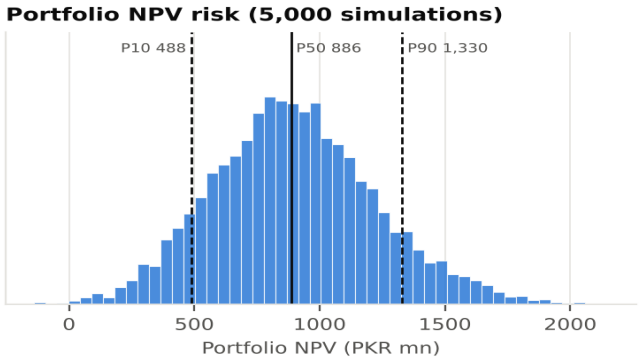
Fund **18 of 38** candidate projects, committing **PKR 2,675 mn** against a five-year budget of PKR 2,760 mn. The portfolio returns an expected NPV of **PKR 1,648 mn** (benefit-cost ratio 1.68x, simple payback 7.0 years from plan start).

Why this set, not a ranked list. The selection and start years were chosen by a mixed-integer optimisation that respects every year's budget, mutually exclusive alternatives, prerequisite projects and regulatory deadlines at the same time. It delivers **+4.8%** more NPV than funding projects in order of NPV, and **+4.4%** more than ranking by NPV per rupee of capex, using the same budget.

Risk. Across 5,000 Monte Carlo simulations of cost overruns, benefit shortfalls, commissioning delays and a shared energy-price shock, portfolio NPV ranges from PKR 488 mn (P10) to PKR 1,330 mn (P90), and is positive in **100%** of outcomes. The single largest driver is **benefit realisation**, which swings NPV by PKR 1,622 mn.

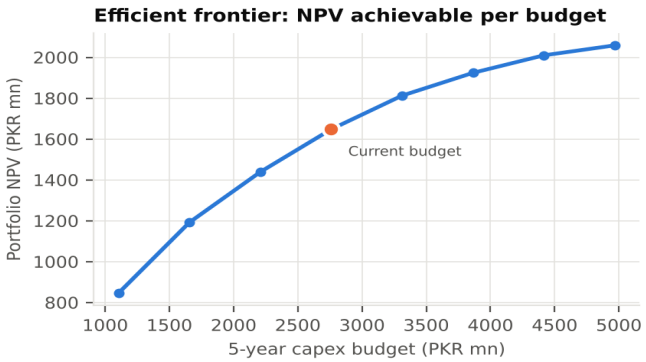
Compliance. 2 regulatory project(s) (P037, P038) are NPV-negative but mandatory; they are funded within their deadlines and their cost is absorbed by the rest of the portfolio.

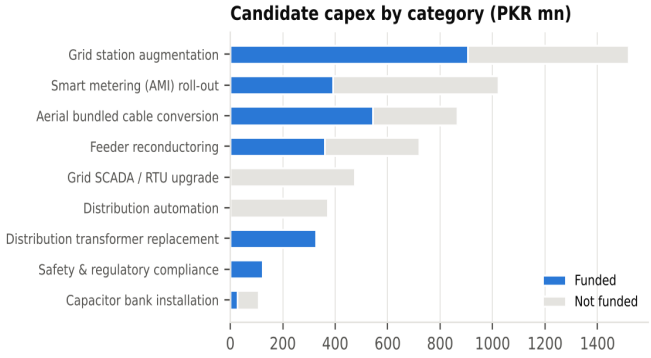




Recommended portfolio

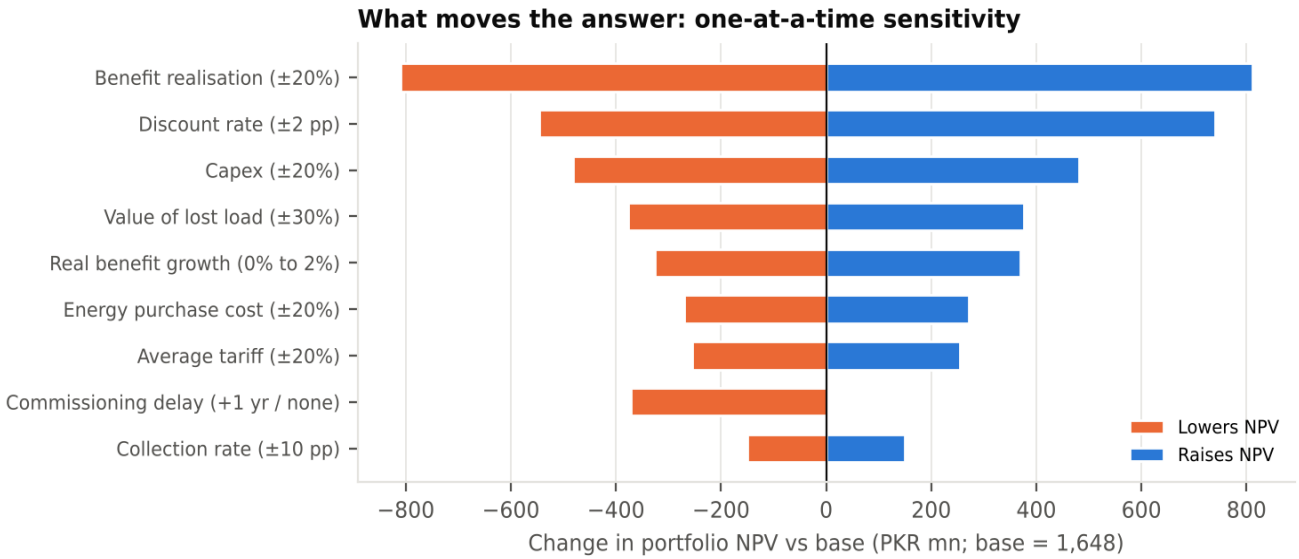
ID	Project	Start	Capex	NPV	IRR	Payback	P(NPV>0)	Risk
P005	Reconductor Feeder 180 (East)	2027	176	285	31%	3.3 y	100%	low
P001	Reconductor Feeder 177 (North)	2027	104	176	32%	3.2 y	100%	low
P007	Reconductor Feeder 684 (West)	2027	78	94	26%	3.9 y	100%	low
P015	Transformer replacement batch 5 (South)	2027	60	70	27%	3.8 y	100%	low
P011	Transformer replacement batch 1 (East)	2027	46	69	31%	3.3 y	100%	low
P020	ABC conversion high-loss area 4 (Central)	2028	193	219	26%	4.0 y	87%	high
P022	ABC conversion high-loss area 6 (South)	2028	158	76	18%	6.1 y	46%	high
P014	Transformer replacement batch 4 (North)	2028	67	68	25%	4.1 y	100%	low
P012	Transformer replacement batch 2 (West)	2028	68	51	22%	4.7 y	100%	low
P038	Earthing & LT rehabilitation (West)	2028	47	-34	-3%	-	0%	low
P037	Earthing & LT rehabilitation (South)	2028	77	-56	-2%	-	0%	low
P035	Grid station augmentation (West)	2029	536	313	18%	6.9 y	66%	high
P025	Smart metering phase 3 (West)	2029	390	185	20%	5.2 y	77%	medium
P034	Grid station augmentation (East)	2029	368	141	16%	7.6 y	41%	high
P013	Transformer replacement batch 3 (Central)	2029	32	35	26%	4.0 y	100%	low
P019	ABC conversion high-loss area 3 (West)	2031	192	132	21%	4.9 y	63%	high
P016	Transformer replacement batch 6 (East)	2031	53	39	22%	4.7 y	100%	low
P004	Capacitor bank Feeder 789 (South)	2031	26	29	30%	3.4 y	100%	low





The frontier shows the NPV the optimiser can deliver at each budget level. Where it flattens, extra budget buys little; where it is steep, a budget increase is itself a high-return decision.

Sensitivity



Projects not funded

ID	Project	Capex	NPV	P(NPV>0)	Reason not funded
P029	Feeder automation cluster 1 (North)	108	147	100%	Prerequisite not worth funding in window
P030	Feeder automation cluster 2 (North)	75	81	100%	Prerequisite not worth funding in window
P036	Grid station augmentation (North)	613	78	10%	Outranked within budget
P017	ABC conversion high-loss area 1 (South)	116	50	40%	Outranked within budget
P010	Reconductor Feeder 487 (North)	117	36	89%	Outranked within budget
P032	Feeder automation cluster 4 (East)	43	36	99%	Prerequisite not worth funding in window
P009	Reconductor Feeder 492 (Central)	116	27	80%	Outranked within budget
P002	Capacitor bank Feeder 177 (North)	31	18	99%	Alternative on same asset selected
P018	ABC conversion high-loss area 2 (East)	120	16	12%	Outranked within budget
P008	Capacitor bank Feeder 684 (West)	34	14	95%	Alternative on same asset selected
P023	Smart metering phase 1 (Central)	324	12	14%	Outranked within budget
P006	Capacitor bank Feeder 180 (East)	16	10	99%	Alternative on same asset selected
P033	Feeder automation cluster 5 (East)	52	2	10%	Prerequisite not worth funding in window
P031	Feeder automation cluster 3 (Central)	92	1	6%	Prerequisite not worth funding in window
P003	Reconductor Feeder 789 (South)	127	-2	29%	Negative NPV
P021	ABC conversion high-loss area 5 (North)	84	-10	1%	Negative NPV
P024	Smart metering phase 2 (South)	308	-24	5%	Negative NPV
P028	Grid SCADA upgrade (East)	124	-86	0%	Negative NPV
P026	Grid SCADA upgrade (North)	216	-98	0%	Negative NPV
P027	Grid SCADA upgrade (Central)	133	-99	0%	Negative NPV

Assumptions

Assumption	Value	Unit	Meaning
avg_tariff	45	PKR/kWh	Average billed tariff; values theft recovered.
base_year	2027	year	First year of the capex plan (plan year index 0).
benefit_growth	0.01	real, p.a.	Real growth of energy-linked benefits with load growth.
collection_rate	0.85	share	Share of newly billed energy actually collected.
discount_rate	0.12	real, p.a.	Real pre-tax WACC used to discount all cash flows.
energy_price_volatility	0.12	sigma	Lognormal volatility of the shared energy-price factor in Monte Carlo.
energy_purchase_cost	28	PKR/kWh	Average cost of energy purchased; values technical losses avoided.
ens_societal	1	flag	1 = value outages at VoLL (societal view); 0 = utility lost margin only.
voll	120	PKR/kWh	Value of lost load - societal cost of 1 kWh not served.

Methodology and limitations

Benefits are derived from physical drivers: technical losses avoided are valued at energy purchase cost; recovered theft at tariff times collection rate; avoided outages at the value of lost load (societal view; switchable to utility margin); plus avoided maintenance cost. All values are real, discounted at a real WACC.

Metrics: NPV, IRR, simple and discounted payback, benefit-cost ratio, profitability index (NPV / PV capex) and equivalent annual value.

Optimisation: binary choice of project and start year, maximising NPV discounted to plan start, subject to annual budgets, mandatory projects with deadlines, mutually exclusive alternatives and prerequisites. Solved with CBC via PuLP. All capex must fall inside the planning window.

Risk: triangular capex and benefit multipliers and discrete delay probabilities by risk class; one lognormal energy-price factor shared across projects (systematic risk). Selection can also be optimised on mean or P10 NPV for a risk-averse plan.

Limitations: synthetic data; no taxes, financing structure or salvage value; projects are assumed independent except through explicit constraints and the shared price factor; deferral shifts a project's whole cash flow without changing its benefits.